GRouNdScale Top-k Causal-Graph Scaling Report

Auto-generated from `publication/tf-k-experiment` run logs

October 16, 2026

1 Overview

This report analyses the 15 completed runs of the publication top-k experiment. The GRN parameter k is the number of top-ranked transcription factors retained per gene (TFs per gene), and k is the independent variable throughout the scaling analysis, replacing imposed-edge count. Imposed edges are shown only as a descriptive column in Table 1. Logs are taken from Slurm job `35270928`, which matches the current `results/` directory. The older job `34606270` and `results-old/` are excluded.

Observed dimensions. The HVG and target-gene counts are constant across runs, but the TF and gene counts are not perfectly fixed: the $k = 1$ run has 194 TFs and 2,997 genes, while all other runs have 197 TFs and 3,000 genes. This is reported as observed; the Notes section includes a sensitivity check that excludes $k = 1$.

2 GRN Characteristics

Table 1: GRN statistics for each top-k setting. Imposed edges are a derived descriptive quantity and are not used as the independent variable.

k	TFs	Targets	Genes	TF/Target (%)	Imposed Edges	Possible Edges	Density (%)
1	194	2803	2997	6.92	2,803	543,782	0.5
2	197	2803	3000	7.03	5,606	552,191	1.0
3	197	2803	3000	7.03	8,409	552,191	1.5
4	197	2803	3000	7.03	11,212	552,191	2.0
5	197	2803	3000	7.03	14,015	552,191	2.5
6	197	2803	3000	7.03	16,818	552,191	3.0
7	197	2803	3000	7.03	19,621	552,191	3.6
8	197	2803	3000	7.03	22,424	552,191	4.1
9	197	2803	3000	7.03	25,227	552,191	4.6
10	197	2803	3000	7.03	28,030	552,191	5.1
11	197	2803	3000	7.03	30,833	552,191	5.6
12	197	2803	3000	7.03	33,636	552,191	6.1
13	197	2803	3000	7.03	36,439	552,191	6.6
14	197	2803	3000	7.03	39,242	552,191	7.1
15	197	2803	3000	7.03	42,045	552,191	7.6

3 Training Step Times

Table 2: Average final-step times (ms) for the causal controller (first phase) and causal generator (second phase).

k	Controller (ms)	Generator (ms)
1	39	41
2	36	45
3	39	48
4	38	54
5	32	74
6	27	73
7	40	84
8	40	105
9	41	119
10	36	124
11	35	137
12	38	153
13	18	176
14	27	191
15	36	220

Table 3: Expected duration for 200,000 controller steps and 500,000 generator steps.

k	Controller	Generator	Total
1	2.2 h	5.7 h	7.9 h (0.33 d)
2	2.0 h	6.2 h	8.2 h (0.34 d)
3	2.2 h	6.7 h	8.8 h (0.37 d)
4	2.1 h	7.5 h	9.6 h (0.40 d)
5	1.8 h	10.3 h	12.1 h (0.50 d)
6	1.5 h	10.1 h	11.6 h (0.48 d)
7	2.2 h	11.7 h	13.9 h (0.58 d)
8	2.2 h	14.6 h	16.8 h (0.70 d)
9	2.3 h	16.5 h	18.8 h (0.78 d)
10	2.0 h	17.2 h	19.2 h (0.80 d)
11	1.9 h	19.0 h	21.0 h (0.87 d)
12	2.1 h	21.2 h	23.4 h (0.97 d)
13	1.0 h	24.4 h	25.4 h (1.06 d)
14	1.5 h	26.5 h	28.0 h (1.17 d)
15	2.0 h	30.6 h	32.6 h (1.36 d)

4 Parameter Counts

k	Generator (excl. CC)	Controller	Critic	Labeler	Anti-Lab.	Gen. Opt.	Crit. Opt.	Lab. Opt.	Anti-Lab. Opt.
1	0.20M	3.77M	3.73M	14.02M	14.02M	0.28M	11.18M	42.04M	42.04M
2	0.38M	3.77M	3.73M	14.03M	14.03M	0.49M	11.19M	42.05M	42.05M
3	0.61M	3.77M	3.73M	14.03M	14.03M	0.75M	11.19M	42.05M	42.05M
4	0.89M	3.77M	3.73M	14.03M	14.03M	1.06M	11.19M	42.05M	42.05M
5	1.21M	3.77M	3.73M	14.03M	14.03M	1.42M	11.19M	42.05M	42.05M
6	1.59M	3.77M	3.73M	14.03M	14.03M	1.83M	11.19M	42.05M	42.05M
7	2.02M	3.77M	3.73M	14.03M	14.03M	2.30M	11.19M	42.05M	42.05M
8	2.50M	3.77M	3.73M	14.03M	14.03M	2.81M	11.19M	42.05M	42.05M
9	3.03M	3.77M	3.73M	14.03M	14.03M	3.37M	11.19M	42.05M	42.05M
10	3.61M	3.77M	3.73M	14.03M	14.03M	3.99M	11.19M	42.05M	42.05M
11	4.24M	3.77M	3.73M	14.03M	14.03M	4.65M	11.19M	42.05M	42.05M
12	4.92M	3.77M	3.73M	14.03M	14.03M	5.36M	11.19M	42.05M	42.05M
13	5.65M	3.77M	3.73M	14.03M	14.03M	6.13M	11.19M	42.05M	42.05M
14	6.44M	3.77M	3.73M	14.03M	14.03M	6.95M	11.19M	42.05M	42.05M
15	7.27M	3.77M	3.73M	14.03M	14.03M	7.81M	11.19M	42.05M	42.05M

Table 4: Parameter counts for each component and optimiser, from `params-size.json`. The generator column excludes the causal controller, which is a submodule of the stored `gen_state_dict` and is shown separately.

5 GPU Memory

Table 5: Peak GPU memory (MiB) by training phase, from the monitoring CSV using the logged phase windows.

k	Controller (MiB)	Generator (MiB)
1	6,231	7,621
2	6,159	7,751
3	6,207	7,835
4	6,159	7,935
5	6,207	8,005
6	6,207	8,091
7	6,207	8,175
8	6,159	8,193
9	6,207	8,251
10	6,207	8,361
11	6,159	8,481
12	6,159	8,619
13	6,207	8,735
14	6,207	8,869
15	6,159	8,855

6 Fits in k

Let k denote the number of TFs per gene. Ordinary least-squares fits over all 15 runs give:

$$\widehat{P}_{\text{gen}}(k) = 5.045 \times 10^5\, k - 1.065 \times 10^6 \text{ parameters}, \quad R^2 = 0.9645, \tag{1}$$

$$\widehat{M}_{\text{gen}}(k) = 87.95\, k + 7548 \text{ MiB}, \quad R^2 = 0.9861, \tag{2}$$

$$\widehat{t}_{\text{gen}}(k) = 12.47\, k + 9.857 \text{ ms/step}, \quad R^2 = 0.9629. \tag{3}$$

Quadratic fits, which capture the curvature visible in the data, are:

$$\widehat{P}_{\text{gen},2}(k) = 2.523 \times 10^4\, k^2 + 1.009 \times 10^5\, k + 7.849 \times 10^4 \text{ parameters}, \quad R^2 = 1.0000, \tag{4}$$

$$\widehat{M}_{\text{gen},2}(k) = 1.179\, k^2 + 69.09\, k + 7602 \text{ MiB}, \quad R^2 = 0.9887, \tag{5}$$

$$\widehat{t}_{\text{gen},2}(k) = 0.5815\, k^2 + 3.163\, k + 36.22 \text{ ms/step}, \quad R^2 = 0.9938. \tag{6}$$

Table 6: Goodness of fit in k (in-sample, all 15 runs).

Quantity	Linear R^2	Quadratic R^2	Linear RMSE	Quadratic RMSE
Generator parameters	0.9645	1.0000	418,359.37	0.00
Generator peak memory (MiB)	0.9861	0.9887	45.18	40.74
Generator step time (ms)	0.9629	0.9938	10.57	4.33

7 Graphs

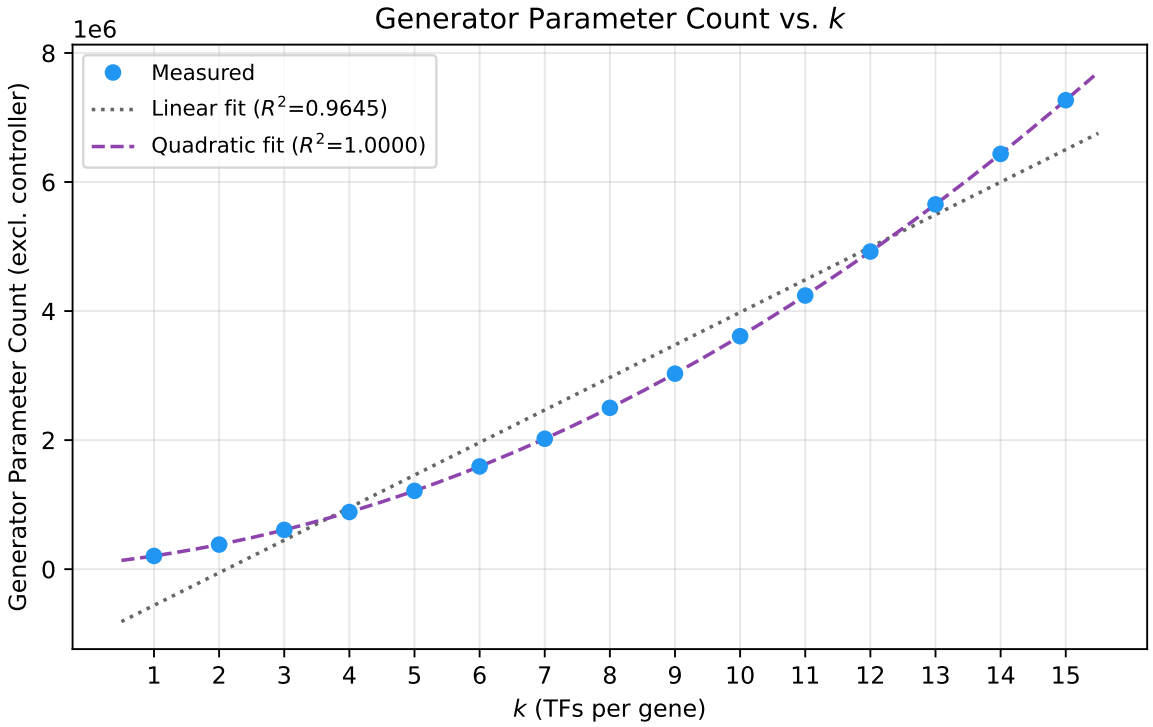


Figure 1: Generator parameter count versus k, with linear and quadratic fits.

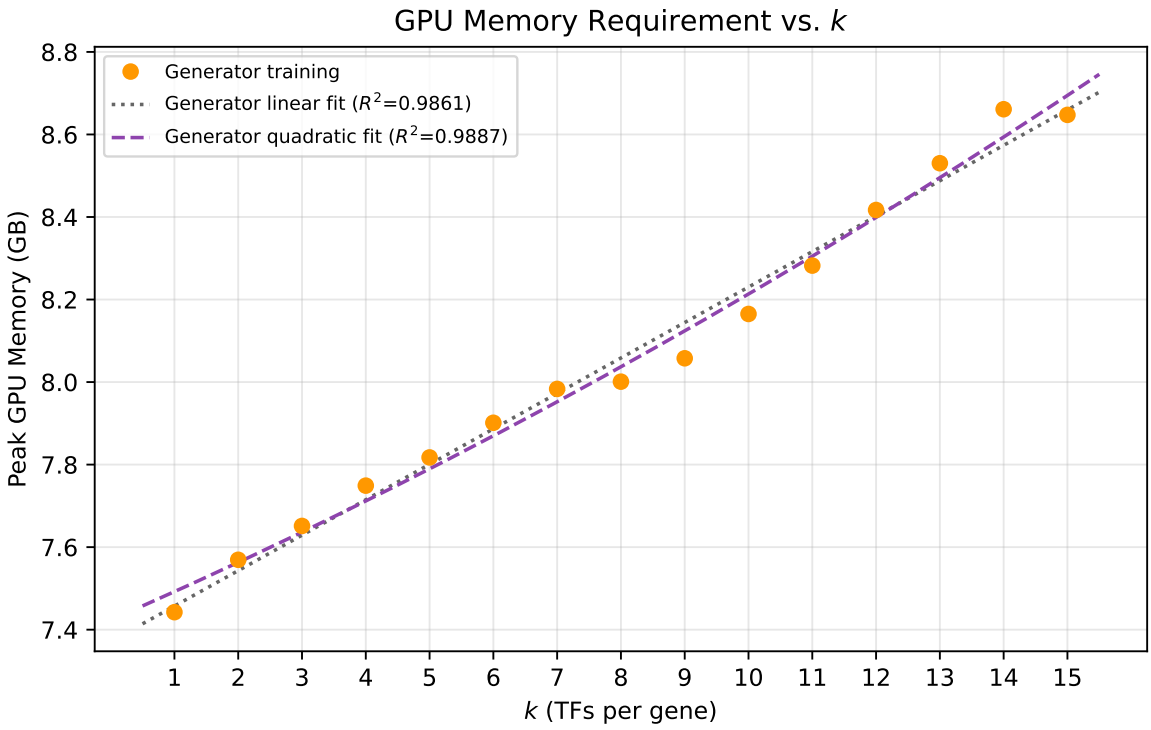


Figure 2: Peak GPU memory versus k, with the generator linear and quadratic fits.

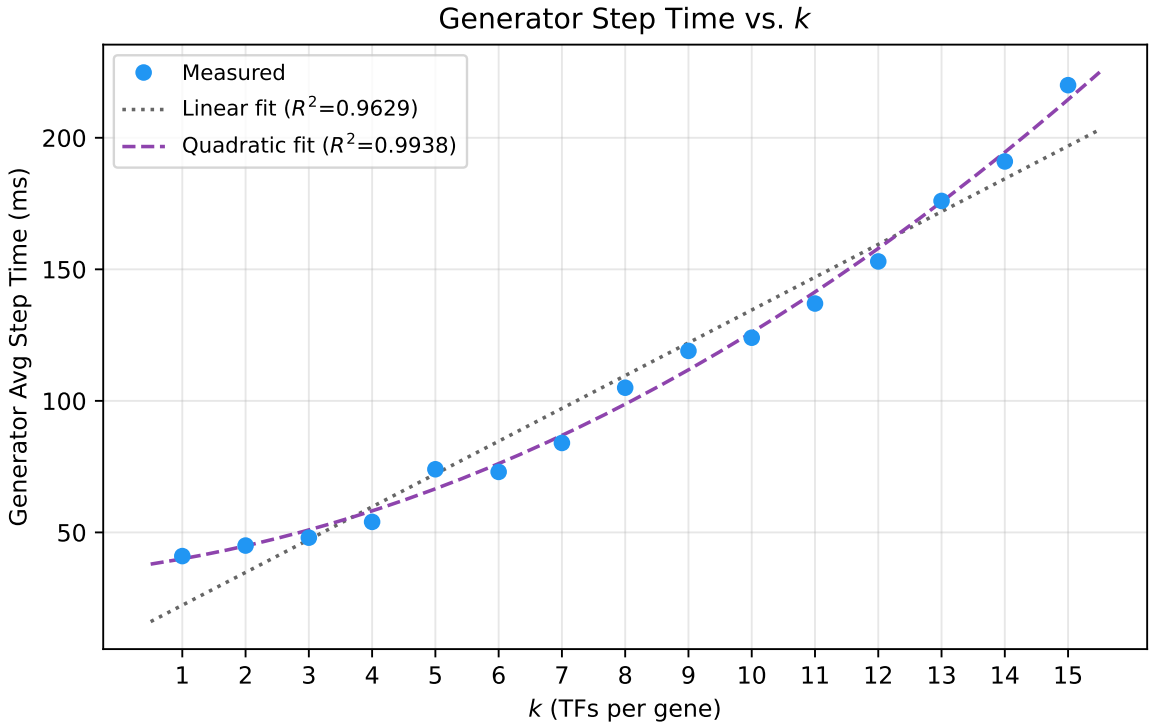


Figure 3: Causal-generator average step time versus k, with linear and quadratic fits.

8 Notes

- Sensitivity to the $k = 1$ point (which has different TF and gene counts): excluding it, the linear/quadratic R^2 are 0.9718/1.0000 for parameters, 0.9829/0.9887 for peak memory, and 0.9697/0.9930 for step time.
- Runs with an out-of-memory error: 0 of 15.
- The linear parameter fit crosses zero at $k \approx 2.1$, so it predicts a negative generator parameter count below that value; the quadratic fit reproduces the measured counts and should be used at small k.
- The TF/target ratio is essentially constant (6.92% at $k = 1$; 7.03% for $k \geq 2$), because the target count is fixed. A ratio-based figure would therefore carry no information and is omitted; the scaling is attributed to k.
- The controller peak memory is flat across k (from 6,159 to 6,231 MiB), as expected because the controller does not depend on k. This supports attributing the generator's growth to k rather than to a phase-window artefact.
- GPU capacity is 81,920 MiB; the largest generator peak is 8,869 MiB at $k = 14$.